Unit 3.01: Gradient Descent & Optimization

Q1. In gradient descent, a weight currently has value w = 20 . If the gradient of the loss is frac{partial L}{partial w} = 8 and the learning rate is eta = 0.2 , find the updated weight w' .

[ ] A: 12.00
[ ] B: 21.60
[x] C: 18.40
[ ] D: 19.80

Solution: Using the gradient descent update rule: w' = w - eta frac{partial L}{partial w} = 20 - 0.28 = 20 - 1.60 = 18.40 .

Q2. In gradient descent, a weight currently has value w = 30 . If the gradient of the loss is frac{partial L}{partial w} = 12 and the learning rate is eta = 0.3 , find the updated weight w' .

[ ] A: 18.00
[ ] B: 33.60
[x] C: 26.40
[ ] D: 29.70

Solution: Using the gradient descent update rule: w' = w - eta frac{partial L}{partial w} = 30 - 0.312 = 30 - 3.60 = 26.40 .

Q3. In gradient descent, a weight currently has value w = 60 . If the gradient of the loss is frac{partial L}{partial w} = 24 and the learning rate is eta = 0.6 , find the updated weight w' .

[ ] A: 36.00
[ ] B: 74.40
[x] C: 45.60
[ ] D: 59.40

Solution: Using the gradient descent update rule: w' = w - eta frac{partial L}{partial w} = 60 - 0.624 = 60 - 14.40 = 45.60 .

Q4. Consider the scalar loss function Lw = 2w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 2 .

[ ] A: nabla L2 = 16
[ ] B: nabla L2 = 8
[x] C: nabla L2 = 0
[ ] D: nabla L2 = -4

Solution: Differentiate with respect to w : frac{dL}{dw} = 22w - 8 = 4w - 8 . At w = 2 : nabla L = 42 - 8 = 8 - 8 = 0 .

Q5. Consider the scalar loss function Lw = 3w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 4 .

[x] A: nabla L4 = 16
[ ] B: nabla L4 = 24
[ ] C: nabla L4 = 32
[ ] D: nabla L4 = 4

Solution: Differentiate with respect to w : frac{dL}{dw} = 23w - 8 = 6w - 8 . At w = 4 : nabla L = 64 - 8 = 24 - 8 = 16 .

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Q6. Consider the scalar loss function Lw = 4w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 6 .

[x] A: nabla L6 = 40
[ ] B: nabla L6 = 48
[ ] C: nabla L6 = 56
[ ] D: nabla L6 = 16

Solution: Differentiate with respect to w : frac{dL}{dw} = 24w - 8 = 8w - 8 . At w = 6 : nabla L = 86 - 8 = 48 - 8 = 40 .

Q7. Consider the scalar loss function Lw = 5w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 8 .

[x] A: nabla L8 = 72
[ ] B: nabla L8 = 80
[ ] C: nabla L8 = 88
[ ] D: nabla L8 = 32

Solution: Differentiate with respect to w : frac{dL}{dw} = 25w - 8 = 10w - 8 . At w = 8 : nabla L = 108 - 8 = 80 - 8 = 72 .

Q8. Consider the scalar loss function Lw = 6w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 10 .

[x] A: nabla L10 = 112
[ ] B: nabla L10 = 120
[ ] C: nabla L10 = 128
[ ] D: nabla L10 = 52

Solution: Differentiate with respect to w : frac{dL}{dw} = 26w - 8 = 12w - 8 . At w = 10 : nabla L = 1210 - 8 = 120 - 8 = 112 .

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